Sanket Prajapati

sanket2742003@gmail.com 9904491017 Vadodara, Gujarat
https://linkedin.com/in/sanket-prajapati27 github.com/sanket913

INTRODUCTION

AI & Machine Learning enthusiast currently pursuing M.Tech in Artificial Intelligence & Data Science with hands-on experience in Python, Machine Learning, Large Language Models (LLMs), healthcare AI, research prototyping, and intelligent application development. Experienced in building AI-driven solutions using Streamlit, FastAPI, Scikit-learn, and modern ML frameworks while translating research ideas into practical prototypes. Passionate about AI research, responsible AI, multi-agent systems, and developing real-world AI applications that create meaningful impact.

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI & Machine Learning: Scikit-learn, TensorFlow (Basic), PyTorch, Machine Learning, Deep Learning, LLM, Prompt Engineering , Model Evaluation

Data Science: Pandas, NumPy, Data Cleaning, Data Preprocessing, EDA, Data Visualization

AI Development: Streamlit, REST APIs, Jupyter Notebook, Google Colab, Git, Github, VS Code

Databases: MongoDB **, My**SQL

Research: Literature Review, Research Paper Analysis, Technical Documentation, Model Benchmarking, Explainable AI

INTERNSHIPS

Machine Learning Intern, ByteXL *May 2024 – Aug 2024*

- Developed a Machine Learning model for software developer salary prediction using Python and Scikit-learn.
- Collected, cleaned, and preprocessed structured datasets using Pandas to improve data quality and model performance.
- Performed exploratory data analysis, feature engineering, and comparative model evaluation to identify meaningful patterns.
- Built an interactive Streamlit application for real-time salary prediction and visualization of analytical insights.
- Documented experimentation process and model performance to support reproducibility and future improvements.

EDUCATION

Master of Technology (AI-Data Science) *Aug 2025 – Present*
Parul University, Vadodara(Gujarat) **CGPA – 9.08/10**

Bachelor of Technology (Information Technology) *Sept 2022 – May 2025*
Parul University, Vadodara(Gujarat) **CGPA – 8.68/10**

Diploma In Engineering (Information Technology) *Aug 2019 – May 2022*
Maharaja Sayajirao University Of Baroda , Vadodara(Gujarat) **Percentage – 85%**

Class 10th - GSEB *Jun 2018 – Mar 2019*
Sabari Vidyalaya, Vadodara(Gujarat) **Percentage – 79.83%**

TECHNICAL PROJECTS

AI-Based Mental Fatigue & Burnout Detection Framework

Skills: Python, PyTorch, TensorFlow, Scikit-learn, Transformers, LSTM, SHAP, LIME, Pandas, Streamlit

- Designed an AI-driven framework for continuous detection of mental fatigue and burnout using smartphone behavioral signals such as screen time, typing patterns, application switching frequency, and night-time device usage.
- Developed personalized behavioral baseline modeling to identify abnormal usage patterns and improve early fatigue detection.
- Built machine learning and deep learning models including LSTM and Transformer architectures for behavioral sequence analysis.
- Applied Explainable AI (SHAP & LIME) to provide transparent predictions and visualize feature importance for responsible AI decision making.
- Developed an interactive healthcare dashboard using Streamlit to monitor burnout risk, behavioral trends, and personalized insights.
- Focused on privacy-aware and non-invasive digital health monitoring using behavioral data instead of sensitive medical records.

RouteWise – AI-Powered Travel Analytics Platform

Skills: React.js, TypeScript, Node.js, Express.js, MySQL, Gemini AI, Dashboard Development

- Developed a travel analytics platform that enables users to analyze trip budgets, optimize travel routes, and generate personalized travel recommendations using structured trip data.
- Designed and implemented data processing workflows to organize travel information, calculate budget estimates, and generate actionable travel insights.
- Built an interactive analytics dashboard to visualize travel expenses, itinerary details, and recommendation results for improved decision making.
- Integrated Gemini AI to generate travel summaries, itinerary recommendations, and contextual insights based on user preferences.
- Collaborated with a two-member team using Git and GitHub to deliver a functional solution during the Odoo × Parul University Hackathon 2026.

Software Developer Salary Prediction

Skills: Python, SQL, Pandas, Scikit-learn, Streamlit

- Collected, cleaned, and preprocessed structured salary datasets using Python and Pandas.
- Performed exploratory data analysis to identify relationships between skills, experience, and salary trends.
- Developed a machine learning model to predict software developer salaries using Scikit-learn.
- Built an interactive Streamlit dashboard to visualize predictions and analytical insights.

PUBLICATIONS

Transparent and Accountable Multimodal Generative AI for Speech Technology

IEEE Conference Publication | 2026

- Published IEEE research proposing the **Hybrid Multimodal Speech Accountability Framework (HMSAF)** for developing transparent, privacy-preserving, and accountable generative speech AI systems.
- Integrated Explainable AI (XAI), multilingual watermarking, adversarial robustness, and voice anonymization to improve transparency, security, and authenticity verification.
- Evaluated the proposed framework using benchmark datasets including **ASVspoof 2025**, **VoicePrivacy 2024**, and **MMP Bench** to assess deepfake detection, privacy preservation, and robustness.
- Addressed challenges related to AI accountability, responsible AI, explainability, and trustworthy deployment of multimodal speech generation systems.

ACHIEVEMENTS

Odoo × Parul University Hackathon 2026
Vadodara Hackathon 4.0
Parul Lets Hack
AI / ML HealthCare Hackathon 1.0 (By myOnsite Healthcare)